

NextGenTrip.com

Prompt-Wise Development Playbook

A copy-paste prompt sequence to build the AI-first hotel extranet, onboarding automation, CRS, channel manager, revenue AI, analytics, support, and production readiness modules from the SOW.

Live GitHub Remote	https://github.com/srprogrammerashishgautam-alt/nextgentrip.git
Permanent Local Workspace	D:\github\nextgentrip
Source SOW	C:\Users\Ashish\Downloads\NextGenTrip_SOW.docx
Important Instruction	Do not start development until each prompt is pasted intentionally and reviewed.

How To Use This PDF

- Paste one prompt at a time into Codex or your coding agent.
- After each prompt, review the diff, run tests, commit, and push to the live GitHub repo.
- Do not start the next phase until acceptance checks for the current phase pass.
- Keep secrets out of Git. Use .env examples and service interfaces/mocks until real credentials are available.
- Use feature branches like module/phase-1-foundation or codex/phase-1-foundation, then merge after review.

Project Breakdown

Module	Description
AI Hotel Acquisition Engine	Automatically discovers and scores hotels from Google Maps, OTAs, social media, and directories
Zero-Touch Onboarding	AI-guided onboarding with auto-fill, OCR document processing, e-sign contracts in under 30 minutes
Property Extranet	Full-featured hotel dashboard with AI Copilot for revenue, inventory, and booking management
Channel Manager	Real-time two-way sync with 20+ OTAs including Booking.com, Expedia, MakeMyTrip, Airbnb
CRS (Central Reservation)	Single source of truth for all rates, inventory, and restrictions across channels
Revenue Management AI	Dynamic pricing, demand forecasting, competitor monitoring, and auto rate optimization
PMS Integration Layer	Connectors for Opera, Cloudbeds, eZee, Hotelogix, and custom PMS systems
AI Support System	Chatbot, WhatsApp agent, voice agent, ticket classification, and auto-resolution
Analytics & Reporting	ADR, RevPAR, occupancy, channel mix, forecasting, and hotel health scoring
Specialty Modules	Medical tourism, wedding tourism, and corporate travel dedicated modules

Database Domains From SOW

Schema / Module	Key Tables	Description
hotels	hotels, hotel_locations, hotel_contacts, hotel_amenities, hotel_policies	Master hotel registry - all property data
rooms	room_types, room_categories, room_amenities, bed_types, occupancy_rules	Room inventory definitions and configuration
rates	rate_plans, rate_values, meal_plans, derived_rates, rate_restrictions	Pricing structures, BAR rates, promotions
inventory	inventory_calendar, allotments, stop_sell, min_stay, close_out	Availability calendar, restrictions, allotments
channels	channels, channel_mappings, channel_credentials, channel_logs, sync_queue	OTA connections, room mapping, sync status
bookings	bookings, booking_rooms, booking_guests, booking_payments, modifications	All reservation data, payment, modifications
kyc	kyc_requests, kyc_documents, kyc_verifications, gst_records, pan_records	KYC workflow, document storage, verification
onboarding	onboarding_sessions, onboarding_steps, onboarding_scores, go_live_checks	Onboarding workflow tracking per hotel

Schema / Module	Key Tables	Description
content	hotel_descriptions, room_descriptions, images, image_tags, seo_content	All AI-generated and approved content
contracts	contracts, contract_versions, contract_signatories, e_signatures	Legal agreements, digital signatures, versioning
revenue	revenue_forecasts, competitor_rates, pricing_rules, pricing_logs	Revenue management data and audit trail
pms	pms_connections, pms_mappings, pms_sync_logs, pms_errors	PMS integration configuration and sync history
analytics	daily_stats, channel_stats, forecast_snapshots, health_scores	Pre-aggregated analytics for dashboards
notifications	notification_templates, notification_logs, whatsapp_logs, sms_logs	All outbound communications
support	tickets, ticket_messages, knowledge_base, ai_resolutions	Help desk and AI support system

Master Prompt Rules

For every NextGenTrip implementation prompt: work from D:\github\nextgentrip; keep the live remote as <https://github.com/srprogrammerashishgautam-alt/nextgentrip.git>; inspect existing files before editing; preserve user changes; implement only the requested phase/module; prefer Laravel, Vue/Inertia, PostgreSQL, Redis, queues, Kafka-style events, and Python FastAPI service boundaries as specified in the SOW; mock external providers until credentials are available; add tests for services, APIs, policies, and critical workflows; run relevant checks; commit with the requested message; push only when the user confirms.

P0 - Repository Bootstrap And Ground Rules

Scope

- Use D:\github\nextgentrip as the permanent local workspace.
- Connect to the live GitHub remote and keep all future module work commit-ready.
- Create README, .gitignore, environment examples, branch strategy, and development conventions.

Copy-Paste Prompt

You are working in D:\github\nextgentrip. Do not build application features yet. Prepare the repository for the NextGenTrip AI-first hotel extranet platform. Verify the Git remote is <https://github.com/srprogrammerashishgautam-alt/nextgentrip.git>, create a clean README, .gitignore, environment template, branch naming rules, module delivery checklist, and GitHub push workflow. Do not overwrite any existing local folder without inspecting it first. Commit only setup/documentation files with message: chore: prepare nextgentrip repository workspace.

Acceptance Checks Before Next Prompt

- Remote URL is correct
- No unrelated local files overwritten
- README explains module-by-module workflow

P1 - Foundation, Infrastructure And Core Setup

Scope

- Laravel 12 modular skeleton, auth, RBAC, audit columns, queues, Kafka topics, admin base.
- Terraform/AWS placeholders and CI/CD workflow structure.
- API versioning under /api/v1 and service mesh conventions for Python AI services.

Copy-Paste Prompt

Implement Phase 1 from the NextGenTrip SOW: Foundation, Infrastructure and Core Setup. Build a Laravel 12 / PHP 8.3 modular monolith skeleton with modules for hotels, rooms, rates, inventory, channels, bookings, kyc, onboarding, content, contracts, revenue, pms, analytics, notifications, support, and audit. Add Sanctum/JWT-ready auth structure, magic link and OTP service interfaces, Spatie RBAC roles and permissions, UUID base models, soft deletes, created_by/updated_by auditing, API response envelope {success,data,message,meta}, /api/v1 route grouping, queue names, Kafka topic config, Horizon config, Filament admin shell, CI placeholders, and tests for auth and permission denial. Keep external services behind interfaces and mocks. Commit as: feat: phase 1 foundation setup.

Acceptance Checks Before Next Prompt

- Migrations run
- Auth tests pass
- Unauthorized APIs return 403
- Queue/topic config documented

P2 - AI Hotel Acquisition And Lead Engine

Scope

- Discovery jobs, Google/OTA/OSM adapters, website crawler interface, CRM lead pipeline.
- 10-dimension hotel scoring algorithm and fuzzy deduplication.
- Bulk CSV/XLSX import and invite flow.

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Implement Phase 2 from the NextGenTrip SOW: AI Hotel Acquisition and Lead Engine. Add acquisition module models, migrations, services, jobs, policies, controllers, resources, and tests for hotel discovery, lead storage, fuzzy duplicate detection, 10-dimension scoring out of 100, CRM pipeline stages, follow-up scheduling, bulk CSV/XLSX import, and invite via email/WhatsApp/SMS adapter interfaces. Implement endpoints POST /api/v1/acquisition/discover, GET /api/v1/acquisition/leads, GET /api/v1/acquisition/leads/{id}, POST /api/v1/acquisition/leads/{id}/score, POST /api/v1/acquisition/leads/{id}/invite, POST /api/v1/acquisition/import. Mock Google Maps, Google Business Profile, OpenStreetMap, crawler, and OTA scrapers. Commit as: feat: phase 2 acquisition lead engine.

Acceptance Checks Before Next Prompt

- Scoring totals 100 max
- Duplicate leads are blocked/merged
- Async discovery returns job_id
- Import validates row errors

P3 - Zero-Touch Hotel Onboarding Workflow

Scope

- 9-step onboarding state machine with rollback and SLA tracking.
- Identity, property fetch, content, rooms, images, documents, contracts, rates, channel mapping.
- 25-point go-live checklist and activation gate.

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Implement Phase 3 from the NextGenTrip SOW: Zero-Touch Hotel Onboarding Flow. Build onboarding sessions, steps, state machine transitions, rollback support, SLA timestamps, event emission, and go-live score checks. Implement the 9 steps: identity verification, auto property fetch, AI content creation, AI room setup, image management, document processing, contract agreement, rate/inventory setup, channel mapping/go-live. Add interfaces for GST/PAN verification, Places/crawler/OTA data merge, GPT content, image processing, AWS Textract OCR, DocuSign/DigiLocker signing, and channel sandbox mapping. Implement blocking and warning checklist items and activation only when mandatory checks pass and score target is met. Add feature tests for full onboarding happy path, rollback, and blocked activation. Commit as: feat: phase 3 zero touch onboarding.

Acceptance Checks Before Next Prompt

- All 9 steps represented
- Rollback tested
- Mandatory go-live blockers enforced
- External APIs mocked

P4 - Property Extranet And AI Copilot

Scope

- Partner dashboard, inventory calendar, rate manager, bookings, channels, revenue charts, content, finance, settings.

- Vue 3/Inertia UI with efficient calendar handling.
- Natural-language Copilot command parser and function execution approvals.

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Implement Phase 4 from the NextGenTrip SOW: Property Extranet and AI Copilot. Build the hotel partner extranet using Vue 3 + Inertia with dashboard, inventory manager, rate manager, booking manager, channel manager, revenue dashboard, content manager, finance, and profile/settings screens. Backend APIs must use pre-aggregated analytics for dashboard data and audit every profile/rate/inventory/content change. Add AI Copilot endpoint POST /api/v1/hotels/{hotel}/copilot that parses supported commands into safe function calls, previews risky changes, executes approved commands, and logs the action. Cover commands like rate increase, stop-sell dates, honeymoon package, RevPAR comparison, allotment update, occupancy analysis, and corporate rate creation. Commit as: feat: phase 4 extranet and copilot.

Acceptance Checks Before Next Prompt

- Dashboard avoids live heavy queries
- Calendar handles 365 days x 50 room types
- Copilot actions are audited
- Risky commands require confirmation

P5 - Channel Manager And CRS Engine

Scope

- CRS as source of truth for rates, inventory, restrictions, and bookings.
- OTA adapter pattern for Booking.com, Expedia, Agoda, Airbnb, MakeMyTrip, Goibibo, Yatra, EaseMyTrip, Cleartrip, Trip.com, Hotelbeds, Google Hotel Ads, and Trivago.
- Redis locks, idempotency, retries, dead-letter queues, WebSocket sync status.

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Implement Phase 5 from the NextGenTrip SOW: Channel Manager and CRS Engine. Build CRS models and services for rate plans, rate values, inventory calendar, allotments, stop-sell, min-stay, close-out, bookings, guests, payments, modifications, and channel mappings. All changes must write to CRS first, emit Kafka-style events, then sync through channel adapters. Implement adapter interfaces and sandbox/mock adapters for Booking.com, Expedia, Agoda, Airbnb, MakeMyTrip, Goibibo, Yatra, EaseMyTrip, Cleartrip, Trip.com, Hotelbeds, Google Hotel Ads, and Trivago. Add Redis distributed locks for room-date-hotel inventory, idempotency keys for booking webhooks, exponential retries, dead-letter queue records, channel logs, and real-time sync status. Commit as: feat: phase 5 crs channel manager.

Acceptance Checks Before Next Prompt

- No direct OTA mutation bypasses CRS
- Booking webhooks are idempotent
- Double booking prevented by locks
- Failed syncs retry and alert

P6 - Revenue Management AI And PMS Integration

Scope

- Demand forecasting, dynamic pricing, competitor monitoring, occupancy/cancellation prediction.
- Python FastAPI AI service boundaries.
- PMS adapter layer for Opera, Cloudbeds, eZee, Hotelogix, AxisRooms, custom CSV/webhooks.

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Implement Phase 6 from the NextGenTrip SOW: Revenue Management AI and PMS Integration. Add revenue management models, services, jobs, APIs, and tests for demand forecasts, dynamic pricing guardrails, competitor rates, occupancy predictions, cancellation risk, pricing logs, and daily AI recommendations. Create a Python FastAPI service boundary or stub package for ML endpoints with mocked responses if models are not ready. Add PMS connection, mapping, sync log, and error models with adapter interfaces for Oracle Opera OHIP, Cloudbeds, eZee, Hotelogix, Little Hotelier/SiteMinder, AxisRooms, and generic webhook/CSV. Ensure all PMS data maps into normalized internal room/rate/booking models. Commit as: feat: phase 6 revenue ai pms integration.

Acceptance Checks Before Next Prompt

- Pricing guardrails enforced
- AI service calls are mockable
- PMS sync is bidirectional where required
- Forecast jobs are scheduled

P7 - Analytics, Support AI And Specialty Modules

Scope

- Pre-aggregated analytics, hotel health score, exports.
- AI support chatbot, WhatsApp assistant, voice agent, ticket classification, auto-resolution.
- Medical tourism, wedding tourism, and corporate travel modules.

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Implement Phase 7 from the NextGenTrip SOW: Analytics, Support AI and Specialty Modules. Build daily_stats, channel_stats, forecast_snapshots, health_scores, analytics APIs, exports to Excel/PDF via queues, and KPI calculations for ADR, RevPAR, occupancy, GOPPAR, TRevPAR, channel mix, comparisons, and hotel health score. Add support ticketing, knowledge base, AI resolution records, chatbot/RAG service interface, WhatsApp assistant interface, voice agent interface, and ticket auto-classification. Add specialty modules for medical tourism, wedding tourism, and corporate travel with dedicated models, APIs, and dashboards. Commit as: feat: phase 7 analytics support specialty modules.

Acceptance Checks Before Next Prompt

- Analytics are pre-aggregated
- Exports run in queues
- Support can escalate to humans
- Specialty modules have tests

P8 - Security, Performance And Production Readiness

Scope

- MFA, row-level hotel isolation, encryption, WAF/rate-limit policies, audit logs.
- Load/performance/security testing with k6, Playwright, OWASP ZAP/Snyk placeholders.
- Production runbooks and launch checklist.

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Implement Phase 8 from the NextGenTrip SOW: Security, Performance and Production Readiness. Add MFA/TOTP structure, brute-force lockout, hotel_id row-level isolation checks, encrypted PAN/bank fields, immutable audit logs, privacy deletion workflow, API throttling, security headers, SAST/dependency scan workflow placeholders, k6 load test scripts, Playwright onboarding E2E tests, Lighthouse CI config, health checks, monitoring alerts, backup/restore runbook, incident runbook, and launch readiness checklist. Verify p95 API, dashboard, channel sync, onboarding, query, CDN, and queue lag targets are measurable. Commit as: feat: phase 8 production readiness.

Acceptance Checks Before Next Prompt

- Tenant isolation tested
- Sensitive fields encrypted
- Load/security scripts exist
- Launch checklist complete

API Reference Summary

Endpoint	Request Body	Response
POST /v1/auth/register	{ name, email, mobile, password, property_type }	201: { user, token, refresh_token }
POST /v1/auth/login	{ email, password } or { magic_token }	200: { user, token, refresh_token, permissions[] }
POST /v1/auth/otp/send	{ mobile } or { email }	200: { otp_ref, expires_in }
POST /v1/auth/otp/verify	{ otp_ref, otp_code }	200: { verified: true, token }
POST /v1/auth/refresh	{ refresh_token }	200: { token, refresh_token }
GET /v1/auth/me	Bearer token header	200: { user, roles, permissions, hotels[] }

Queue And Event Architecture

Kafka Topic	Consumer Service	Partitions	Trigger & Action
hotel.created	OnboardingService	12	New hotel record created -> Trigger Step 1 of onboarding workflow
hotel.kyc.submitted	KYCService	12	Documents uploaded -> Start OCR + AI validation pipeline
hotel.activated	ChannelSyncService, NotificationService	6	Hotel goes live -> Push to all mapped channels, send welcome email
rate.updated	ChannelSyncService	24	Rate change in CRS -> Parallel push to all OTAs within 30s
inventory.updated	ChannelSyncService	24	Inventory change -> Push to all channels, update search index
booking.received	BookingService, InventoryService	24	New booking from any channel -> Process, decrement inventory, notify
booking.cancelled	BookingService, InventoryService	24	Cancellation received -> Restore inventory, trigger refund, notify
ai.content.requested	AIContentService (Python)	12	Onboarding Step 3 trigger -> GPT-4o content generation pipeline
ai.images.requested	AllImageService (Python)	12	Onboarding Step 5 trigger -> Download, process, categorize images
ai.pricing.requested	AI PricingService (Python)	6	Nightly trigger per hotel -> ML pricing recommendations
notification.send	NotificationService	12	Any notification event -> Route to email/SMS/WhatsApp/push
analytics.update	AnalyticsService	6	Nightly -> Pre-aggregate all KPIs, update health scores

Testing Strategy

Test Type	Tool / Framework	Coverage Requirement
Unit Tests	PHPUnit + Pest	All service classes, domain models, calculation logic. Min 80% code coverage.
Feature Tests	Laravel HTTP Tests	All API endpoints - happy path + edge cases + auth failures. 100% endpoint coverage.
Integration Tests	PHPUnit + test DB	Channel sync flows, booking processing, inventory management. All state machines.

Test Type	Tool / Framework	Coverage Requirement
E2E Tests	Playwright	Full onboarding flow, extranet key workflows, booking modification. Run on every PR.
Load Tests	k6	5000 concurrent users, 30-minute sustained load. All p95 latency targets must pass.
Security Tests	OWASP ZAP + Snyk	OWASP Top 10 automated scan. Dependency vulnerability scan. Block deployment on critical CVEs.
AI Model Tests	pytest + custom eval	Content quality scoring, OCR accuracy (>95%), room detection precision (>90%).
Channel Sandbox Tests	OTA Sandbox APIs	Full booking lifecycle on Booking.com, Expedia, MakeMyTrip test environments.

Delivery Milestones

#	Task	Assignee	Acceptance Criteria
1	Provision AWS infrastructure using Terraform IaC - VPC, subnets, RDS, ElastiCache, ECS cluster, S3 buckets, CloudFront distribution	DevOps	All resources running, costs within budget, Terraform state stored in S3
2	Create Laravel 12 project with modular structure - separate modules for hotels, bookings, channels, kyc, revenue, analytics	Backend Lead	Module auto-discovery works, each module has own migrations, routes, controllers
3	Implement JWT + Sanctum hybrid auth with refresh tokens, magic login links (signed URLs expiring in 15 min), and OTP (SMS + email)	Backend Dev	Auth tests pass, token refresh works, magic link flow tested end-to-end
4	Build Spatie Permission-based RBAC with 8 roles and 60+ permissions granular to each module and action	Backend Dev	All role-permission combinations tested, API returns 403 on unauthorized access
5	Create all base database migrations with proper indexing, foreign keys, and audit columns (created_at, updated_at, deleted_at, created_by)	Backend Dev	All migrations run without error, indexes validated with EXPLAIN ANALYZE
6	Set up GitHub Actions workflow: lint -> test -> build Docker image -> push to ECR -> deploy to ECS (dev on every push, prod on release tag)	DevOps	Pipeline runs in under 8 minutes, failed tests block deployment
7	Configure Laravel Horizon for queue management with separate queues: default, high, critical, ai-jobs, channel-sync, notifications	Backend Dev	Horizon dashboard accessible, queue depths visible, failed jobs auto-retry 3x
8	Set up Kafka topics for all async events with consumer groups for each service	Backend Dev	All topics created, dead-letter queues configured, messages retained for 7 days

Final Success Metrics

KPI	Target	Measurement Method
Hotel onboarding automation rate	>= 95%	Steps completed without human intervention / total steps
KYC auto-verification rate	>= 90%	AI-verified KYC / total KYC submissions
Room auto-detection accuracy	>= 90%	AI-detected rooms matching manual setup / total hotels
Content auto-generation acceptance rate	>= 85%	AI content used without major edits / total generated
Channel mapping automation rate	>= 90%	Auto-mapped rooms / total mapping tasks
Hotel go-live time (median)	< 30 minutes	Median time from account creation to channel activation
Channel sync latency (rate push p95)	< 30 seconds	OTA API push completion timestamp - CRS update timestamp
Platform uptime	>= 99.99%	AWS Route53 health check + Pingdom external monitoring
API response time p95	< 200ms	CloudWatch + k6 load test reports
Hotels supported at scale	>= 100,000	Load test simulating 100K hotels with active sync